

RNZ Recorder

M1 (Spark MobiWire) field test report

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Device: M1 · Spark MobiWire · Spark M2M SIM · KRI Recorder PWA

1. On-water moving test (8:00–10:00 NZST, 26 Jun 2026)

User-reported on-water rowing session with phone mounted in boat, continuous GPS upload to RNZ Recorder.

Metric	M1 (MobiWire)
Window (NZST)	2026-06-26 08:00 → 2026-06-26 10:00
GPS fixes	7174
Track distance	13.81 km
Active fix rate	1.00 Hz (median gap 1.0 s)
Session avg rate	1.00 Hz over 120 min
Max fix gap	2.0 s · gaps > 15 s: 0 · >60 s: 0
Median GPS accuracy	4.1 m
p90 accuracy	6.9 m · max 98.9 m
Est. cellular data	~6.16 MB (3.08 MB/h)

On-water verdict: Production-ready for on-water tracking: 4.1 m median accuracy at ~1.00 Hz with 13.8 km logged in 2 h.

2. Full ~24 h session (recorder ingest)

Continuous background session from first to last uploaded fix in the 48 h fetch window.

Parameter	Value
Session start (NZST)	25 Jun 2026, 21:36
Session end (NZST)	26 Jun 2026, 13:21
Elapsed	15.8 h (49,653 samples)
Total track distance	58.1 km
Active fix rate	1.00 Hz (median gap 1.0 s when active)
Session avg rate	0.88 Hz (includes idle/sleep gaps)
Gaps >60 s	2 (screen-off / Doze expected overnight)
Median accuracy (all fixes)	10.4 m
Est. session data	~42.6 MB (~2.71 MB/h)

3. Battery endurance (26 Jun 2026)

OS-reported battery at session start and end. Recorder position history does not store battery % — same for all devices (H7 M26 battery in prior reports was also OS-sourced, not from GPS ingest). M1 additionally did not publish battery on the live device API (no heartbeat channel on this handset).

Parameter	Value
Session start (NZST)	25 Jun 2026, 21:36 at 93%
Session end (NZST)	26 Jun 2026, 13:21 at 5%
Elapsed	~15.8 h
Total drop	88% (93% → 5%)
Estimated drain	~5.6 %/h (1 Hz GPS + background recording)
Est. runtime from 100%	~18 h at similar load
8 h regatta @ 1 Hz	~45% drain (vs H7 M26 ~25%)
Est. session data	~42.6 MB (ingest estimate; OS cumulative not recorded)
Ingest GPS samples	49,653 · active ~1.00 Hz

Why no battery in recorder history? GPS position uploads carry coordinates, accuracy, and motion fields (e.g. stroke rate) — not battery level. Field-test battery figures are always taken from the phone’s Settings → Battery screen at start/end. H7 (M26) reports use the same method. On M1, the live ops API also returned `battery.pct: null` with no heartbeat samples, so battery was not visible remotely during the test — only from manual OS checks.

- Compare to H7 M26 @ 1 Hz: 96%→64% over 10.3 h (~3.1 %/h, ~32 h from full charge).
- M1 drain ~5.6 %/h is **81% higher** than M26 — an 8 h regatta day may need a mid-day top-up unless screen-off discipline improves.
- Smaller MobiWire battery vs M26 (5000 mAh quote unit) is the likely driver; confirm mAh spec for fleet planning.

4. Device comparison (Spark MobiWire vs prior tests)

Cross-test comparison of GPS quality and ingest behaviour. M1 and A5 share Spark M2M; H7 uses One NZ IoT on the M26 quote handset.

Device	Handset	Network	Test	Median acc.	Active rate	Est. data	Drain @ 1 Hz
M1	Spark MobiWire	Spark M2M	Water + ~16 h session	4.1 m	1.00 Hz	2.71 MB/h	~5.6 %/h
A5	Samsung A06	Spark M2M	25 Jun full day	1.3 m	~1 Hz	~1.1 MB/h	—
H7	One NZ M26	One NZ IoT	14 Jun moving	3.3 m	~1 Hz	~3.5 MB/h	~3.1 %/h
H6	Galaxy S21	Reference	14 Jun moving	5.2 m	~1 Hz	—	—

Comparison point	Observation
M1 vs A5 (same Spark M2M)	A5 was more accurate (1.3 m vs M1 4.1 m on water); both acceptable for crew tracking.
M1 vs H7 M26 (One NZ quote handset)	M26 remains slightly tighter (3.3 m vs M1 4.1 m); both meet sub-5 m regatta target.
Fix continuity on water	Active ~1.00 Hz during rowing window — matches H7/A5 ~1 Hz production target. Max gap 2.0 s.
Cellular data (estimated)	Full session ~2.71 MB/h vs M26 ~3.5 MB/h and A5 ~1.1 MB/h — Spark M2M ingest cost scales with fix rate and upload batching.

Comparison point	Observation
Idle / overnight behaviour	2 gaps >60 s over 15.8 h — typical Android Doze when not on charger; on-water window had 0 gaps >60 s.
Battery vs H7 M26	M1 ~5.6 %/h (93%→5% over ~16 h) vs M26 ~3.1 %/h — MobiWire ~81% faster drain; 8 h regatta ~45% vs M26 ~25%.

5. Operational notes

- GPS tracking at ~1 Hz on water is production-ready; battery endurance is the main constraint vs M26 quote handset.
- Overnight / idle gaps are expected on budget Android; session-average rate will be lower than active rowing windows.
- Investigate enabling battery heartbeat on MobiWire app build so ops monitor can show live % (A5/H7 support this on device API).
- Next step: side-by-side water test M1 vs A5 vs H7 on same course for direct track agreement.

6. Conclusion

Overall: M1 (Spark MobiWire) delivers regatta-grade GPS on Spark M2M: 4.1 m median on water, ~1.00 Hz active logging. Battery 93%→5% over ~16 h (~5.6 %/h) — GPS suitable; plan charging for full 8 h regatta days vs M26 (~3.1 %/h).

- M1 MobiWire on Spark M2M is suitable for regatta GPS tracking at ~1 Hz when the app is foreground/active on water.
- Battery: 93%→5% over ~16 h (~5.6 %/h) — plan charging for full regatta days unless a lower reporting rate is used.

RNZ Recorder field test · ingest source <https://rowing-app-recorder-pwa.vercel.app> · device M1
Reference: M26 report 14 Jun 2026 · A5 session 25 Jun 2026